

Datatype Mapping: CoreDataType / CLR / SQL Server / Oracle / PostgreSQL

Schema-conversion reference — defaults mirror the TflOmniDb provider type mappers.

CoreDataType	CLR / .NET	SQL Server	Oracle	PostgreSQL	Notes
Boolean & integer					
Boolean	bool	BIT	NUMBER(1,0)	BOOLEAN	Oracle <23c: NUMBER(1) 0/1; 23c+ has BOOLEAN
(none)	byte	TINYINT	NUMBER(3,0)	SMALLINT	TINYINT is unsigned 0-255; no byte CoreDataType
Int16	short / Int16	SMALLINT	NUMBER(5,0)	SMALLINT	
Int32	int / Int32	INT	NUMBER(10,0)	INTEGER	
Int64	long / Int64	BIGINT	NUMBER(19,0)	BIGINT	
(none)	long (identity)	INT IDENTITY	GENERATED AS IDENTITY	GENERATED AS IDENTITY / serial	Identity = KeyGeneration strategy, not a type
Exact numeric					
Decimal	decimal	DECIMAL(p,s)	NUMBER(p,s)	NUMERIC(p,s)	NUMERIC == DECIMAL
Decimal	decimal	NUMERIC(p,s)	NUMBER(p,s)	NUMERIC(p,s)	Oracle NUMBER max precision 38
Money	decimal	MONEY	NUMBER(19,4)	NUMERIC(19,4)	Avoid PG MONEY (locale); canonical NUMERIC(19,4)
(none)	decimal	SMALLMONEY	NUMBER(10,4)	NUMERIC(10,4)	Map to Decimal/Money; range ~ +/-214,748.3647
Approximate numeric					
Single	float / Single	REAL	BINARY_FLOAT	REAL	REAL = FLOAT(24), 4-byte
Double	double / Double	FLOAT	BINARY_DOUBLE	DOUBLE PRECISION	FLOAT(53) = 8-byte double; FLOAT(24)->REAL
Character & text					
AnsiString	string	VARCHAR(n)	VARCHAR2(n CHAR)	VARCHAR(n)	Oracle VARCHAR2 <= 4000 bytes (32767 extended)
UnicodeString	string	NVARCHAR(n)	NVARCHAR2(n)	VARCHAR(n)	Unicode; PG is UTF-8 native (no N-prefix)
AnsiStringFixed	string	CHAR(n)	CHAR(n CHAR)	CHAR(n)	Fixed length, space-padded
UnicodeStringFixed	string	NCHAR(n)	NCHAR(n)	CHAR(n)	Unicode fixed length
Clob	string	VARCHAR(MAX)	CLOB	TEXT	Large ANSI text (> 8000 bytes)
NClob	string	NVARCHAR(MAX)	NCLOB	TEXT	Large Unicode text
Clob	string	TEXT (legacy)	CLOB	TEXT	Deprecated -> use VARCHAR(MAX)
NClob	string	NTEXT (legacy)	NCLOB	TEXT	Deprecated -> use NVARCHAR(MAX)
Date & time					
Date	DateTime / DateOnly	DATE	DATE	DATE	Oracle DATE INCLUDES time-of-day (to sec)
Time	TimeSpan / TimeOnly	TIME(p)	INTERVAL DAY TO SECOND	TIME(p)	Oracle has no TIME-of-day type
DateTime	DateTime	DATETIME2(p)	TIMESTAMP(p)	TIMESTAMP(p)	Preferred date+time; p = frac-sec precision
DateTime	DateTime	DATETIME (legacy)	TIMESTAMP(3)	TIMESTAMP(3)	~3.33 ms precision; range 1753-9999

CoreDataType	CLR / .NET	SQL Server	Oracle	PostgreSQL	Notes
DateTime	DateTime	SMALLDATETIME	TIMESTAMP(0)	TIMESTAMP(0)	Minute precision; 1900-2079
DateTimeOffset	DateTimeOffset	DATETIMEOFFSET	TIMESTAMP WITH TIME ZONE	TIMESTAMPTZ	Timezone-aware
Binary					
Binary	byte[]	VARBINARY(n)	RAW(n)	BYTEA	Oracle RAW <= 2000 bytes; beyond -> BLOB
BinaryFixed	byte[]	BINARY(n)	RAW(n)	BYTEA	Fixed; PG has no fixed-length binary
Blob	byte[]	VARBINARY(MAX)	BLOB	BYTEA	Large binary object
Blob	byte[]	IMAGE (legacy)	BLOB	BYTEA	Deprecated -> use VARBINARY(MAX)
Identifier & concurrency					
Guid	Guid	UNIQUEIDENTIFIER	RAW(16)	UUID	Oracle SYS_GUID(); PG native UUID
(none)	byte[] (rowversion)	ROWVERSION	(none)	(xmin sys col)	IsRowVersion flag -> app-managed int version
XML & JSON					
Xml	string / SqlXml	XML	XMLTYPE	XML	Native XML on all three
Json	string	JSON / NVARCHAR(MAX)	JSON (21c+)/CLOB	JSONB	PG JSONB preferred; Oracle <21c CLOB + IS JSON
No direct equivalent (redesign)					
(none)	object	SQL_VARIANT	(none / ANYDATA)	(none)	Redesign to a typed column
(none)	(tree)	HIERARCHYID	(none)	LTREE (extension)	Remodel as a materialized path
(none)	(spatial)	GEOGRAPHY / GEOMETRY	SDO_GEOMETRY	geometry/geography	Oracle Spatial / PostGIS extension

General notes

1. CoreDataType is TflOmniDb's provider-neutral type enum (Core/CoreDataType.cs); the per-provider type mappers translate it to the native types in each column. '(none)' marks a SQL Server type with no canonical CoreDataType - map it to the nearest enum value, or set a per-column native override.
2. Per-column overrides exist via the [DbColumn] SqlServerType / OracleType / PostgresType escape hatches for the rare native-type need.
3. Money is deliberately NUMERIC(19,4) on all three engines - PostgreSQL's MONEY type is locale-dependent and unsafe for financial code.
4. Identifier casing: Oracle folds unquoted identifiers to UPPER, PostgreSQL to lower. Quote consistently, or pick one case across the schema.
5. Oracle DATE always carries a time component (to seconds). Use TIMESTAMP for fractional seconds; for pure dates, enforce truncation in the app.
6. Booleans: target Oracle <23c as NUMBER(1) with 0/1 values; Oracle 23c+ and PostgreSQL have a native BOOLEAN.
7. Auto-increment: PostgreSQL and Oracle 12c+ support GENERATED ... AS IDENTITY; older Oracle needs a SEQUENCE + trigger.
8. Row versioning: there is no portable equivalent to SQL Server ROWVERSION. Use an app-managed integer version column (TflOmniDb IsRowVersion) checked in the UPDATE WHERE clause; Oracle ORA_ROWSCN / PostgreSQL xmin are not drop-in replacements.
9. LOBs map (N)VARCHAR(MAX)/VARBINARY(MAX) -> Oracle CLOB/NCLOB/BLOB -> PostgreSQL TEXT/BYTEA.
10. PostgreSQL is UTF-8 native, so NVARCHAR and VARCHAR both collapse to VARCHAR/TEXT; JSONB beats JSON for indexing (GIN); BYTEA covers all binary widths.